Chapter 2: Literature Review

2.1 Introduction

Recent advances in artificial intelligence have reshaped digital media forensics, largely due to the rapid rise of deepfakes. These synthetic videos and images, generated through deep neural architectures, can imitate human appearance and behaviour with increasing precision (He, 2021; Rossler et al., 2019). Although early work treated manipulated media as a relatively contained technical issue, current developments have made detection more difficult. As He (2021) pointed out, the realism achieved by modern generative models disrupts long-standing assumptions about what constitutes trustworthy audio-visual evidence. Rossler et al. (2019) further highlight that detection systems often are behind the pace of generative model improvements, creating a persistent asymmetry between manipulation and forensics. Given this trajectory, a clearer understanding of the underlying deep learning methods is necessary before evaluating why current detection strategies struggle and where they fall short.

2.1.1 Define Deep Learning

Deep learning is defined as a subset of machine learning characterised by the utilisation of multi-layered neural networks. This architectural paradigm draws inspiration from the biological neural networks of the human brain, aiming to model high-level abstractions in data through a deep graph with multiple processing layers (LeCun, 2015). According to Goodfellow et al. (2016), deep learning systems differentiate themselves from conventional machine learning by their ability to learn representations of data with multiple levels of abstraction. While a shallow network might only have one hidden layer, a “deep” architecture utilises a hierarchy of layers to process data through a series of nested non-linear equations. This structure allows the model to function as a universal approximator, capable of modeling complex high-dimensional data patterns that simpler algorithms cannot capture (Hornik, 1989).

2.1.2 Define Deepfake

The term deepfake combines “deep learning” and “fake,” which describes synthetic media generated through deep neural models rather than conventional editing techniques (Westerlund, 2019). Most early systems relied on autoencoders or Generative Adversarial Networks, which made it possible to automate face swapping and expression transfer in a way that traditional manipulation tools could not achieve (Gera, 2018). Although digital alterations long predate deep learning, the specific phenomenon of deepfakes entered mainstream attention in late 2017 when a Reddit user released code that enabled realistic face swaps with very little technical skill, initially leading to widespread non-consensual pornography and, later, to wider use in misinformation and entertainment (Chesney and Cytowic, 2019; Westerlund, 2019). In research contexts, deepfakes are generally defined by their use of deep generative models to produce content that is difficult to distinguish from authentic footage (Cai, 2024). Over time, beyond face swaps, current systems include audio-based voice cloning and multimodal models capable of synchronising generated speech with fabricated lip movements, which introduces additional complexity for forensic detection (Yi et al., 2023). Together, these developments show that deepfakes are not a single technique but a growing family of generative methods with expanding forensic implications.

2.2 Evolution of Deepfakes

2.2.1 Phase 1: Visual Fidelity and Generative Advances

The initial phase of deepfake development focused on face swapping, using simple autoencoders and early Generative Adversarial Networks (GANs) (Dolhansky et al., 2020). These techniques were prone to noticeable flaws, such as affine warping errors, making them easily detectable by early forensic tools (Li et al., 2018). However, the field progressed rapidly with the adoption of improved training strategies such as the Two-Time Scale Update Rule (TTUR), which greatly improved GAN stability and convergence, resulting in more lifelike image generation (Heusel et al., 2017). As generative methods evolved, benchmark datasets such as the Deepfake Detection Challenge (DFDC) (Dolhansky et al., 2020) and Celeb-DF (Li et al., 2019) emerged. These resources marked a transition from low-grade and easily spotted fakes to high-quality videos that accurately replicated real-world visuals, posing significant challenges for detection systems (Li et al., 2019).

2.2.2 Phase 2: "In-the-Wild" Adaptation and Multi-Subject Expansion

A major advancement in deepfake technology was transitioning from controlled laboratory settings to more unpredictable "in-the-wild" environments. Although earlier datasets typically included individual subjects under uniform lighting and framing, newer deepfake datasets like Wild Deepfake (Zi et al., 2021) and FFIW-10K (Zhou et al., 2021) consist of content sourced from the internet or specifically generated to endure real-world challenges such as compression and noise. During this stage, deepfake methods were also advanced, enabling the manipulation of multiple people within the same scene. For example, research by Narayan et al. (2023) using the DF-Platter dataset showed that current techniques can alter multiple faces in a single image, even in challenging scenarios such as occlusion or poor image quality. Similarly, the FFIW-10K dataset highlighted the ongoing difficulty that forensic systems face in detecting fakes in group videos—especially when only some faces have been manipulated (Zhou et al., 2021).

2.2.3 Phase 3: Multimodal and Neural Generation

The latest and arguably most concerning development in deepfake technology is the move from only visual edits to multimodal deepfakes that manipulate both audio and video. Recent studies highlight the importance of inconsistencies between sound and visuals, leading to the creation of datasets such as AV-Deepfake1M (Cai, 2024) and FakeAVCeleb (Yi et al., 2023), which feature coordinated alterations to both video and audio streams. This evolution requires that detection tools now evaluate the synchronisation of lip movements with speech, rather than just looking for visual inconsistencies (Yi et al., 2023). In addition, the generative techniques themselves have progressed beyond traditional GANs. Neural Radiance Fields (NeRF) have been used to create talking-head videos where both viewpoint and audio can be modified (Guo et al., 2021). For audio, generation has advanced from simple concatenation methods to diffusion models, such as NaturalSpeech 2, which enable zero-shot voice synthesis with remarkably accurate intonation and prosody (Shen et al., 2023).

2.3 Documented Real-World Impacts and High Profile Cases

Although early research on deepfakes largely focused on improving synthesis quality and benchmarking detection accuracy, larger concern has been driven by their increasing use in real-world scenarios (Chesney and Cytowic, 2019; Dolhansky et al., 2020; Rossler et al., 2019).

One area of significant impact has been medical misinformation. Investigative reporting has documented numerous cases in which deepfake videos impersonating qualified doctors were used to promote false health advice and fraudulent treatments across social media platforms (ITV News, 2024; Clark, 2025). These videos often exploit the visual credibility of trusted medical professionals, making misinformation difficult for non-expert audiences to identify. Such cases highlight how deepfakes can undermine public trust in expert knowledge, particularly in domains where inaccurate information may cause physical harm (Chesney and Cytowic, 2019; Rossler et al., 2019).

Deepfakes have also been employed in financial fraud and impersonation scams. A notable case involved a multinational engineering firm in which employees were deceived into authorising a large financial transfer after participating in a video call that appeared to feature senior executives but was later confirmed to be AI-generated (Milmo, 2024). Similar incidents have been reported globally, prompting warnings from law-enforcement agencies about the growing use of synthetic audio and video in social engineering attacks (Bragg, 2025). These cases demonstrate how deepfakes can amplify traditional fraud techniques by adding a convincing visual and auditory layer.

In addition to visual manipulation, audio deepfakes have demonstrated significant real-world impact, particularly in the context of fraud and impersonation (Korshunov and Marcel, 2018; Yi et al., 2023; Zhang, 2025). One of the first high-profile cases occurred in 2019, when attackers used a synthetic voice to mimic a company executive to authorise a fraudulent financial transfer, resulting in substantial financial losses (Stupp, 2019). Unlike visual deepfakes, audio-based impersonation can be deployed in real-time through phone calls or voice messages, limiting opportunities for human verification and increasing its effectiveness in social engineering scenarios (Korshunov and Marcel, 2018). Prior research demonstrates that synthetic speech can convincingly replicate speaker identity, enabling attacks that bypass traditional speaker verification and authentication systems (Korshunov and Marcel, 2018; Yi et al., 2023). Taken together, these findings indicate that audio deepfakes pose a parallel threat to trust and identity assurance mechanisms, reinforcing the need for detection approaches that extend beyond visual analysis and address multimodal manipulation (Rossler et al., 2019; Cai, 2024).

Political manipulation represents another area of concern associated with deepfakes, particularly in the context of increased geopolitical tension (Chesney and Cytowic, 2019; Vaccari, 2020). Videos depicting public figures have circulated online, sometimes appearing to show officials making false statements or announcements, thereby creating confusion regarding the authenticity of political communication (Rossler et al., 2019). An example involves a manipulated video of a national leader that was briefly disseminated on social media before being removed by platform moderators (Allyn, 2022). Although such videos do not always achieve their intended persuasive impact, studies suggest that their rapid circulation can take advantage of periods of uncertainty and contribute to decline of confidence in authentic news sources (Vaccari, 2020).

In addition to large-scale misinformation, deepfakes have caused direct personal harm. Investigations of non-consensual synthetic pornography have revealed extensive platforms dedicated to generating explicit content using the likenesses of individuals without consent, disproportionately targeting women (Moore, 2025). These cases underscore the ethical and legal challenges posed by deepfakes, particularly in situations where existing regulatory and legal frameworks struggle to address harms arising from fabricated yet highly realistic media (Chesney and Cytowic, 2019; Westerlund, 2019).

Taken together, these examples illustrate that deepfakes are no longer a purely technical problem confined to academic datasets. Their real-world deployment exposes weaknesses in current detection systems and reinforces the need for reliable, generalisable forensic approaches capable of operating under the unpredictable conditions of online media ecosystems (Rossler et al., 2019; Dolhansky et al., 2020).

2.4 The Dual Dilemma: Creative Application Versus Malicious Threat

Deepfake technologies exemplify a dual-use dilemma, in which the same deep generative models that enable legitimate and creative applications can also be exploited for harmful purposes. Advances in audio and visual synthesis have supported a range of beneficial use cases, including film post-production, automated dubbing, digital de-ageing, and assistive technologies such as personalised text-to-speech systems and voice restoration tools. In these contexts, synthetic media techniques are typically deployed with consent and transparency, and their use is framed as an extension of existing digital production methods (Westerlund, 2019; Yi et al., 2023).

However, the properties that make deepfake technologies attractive for creative and commercial applications, namely high realism, automation, and scalability, also enable malicious misuse. As discussed in Section 2.3, deepfakes have been employed in misinformation campaigns, fraud, impersonation, and non-consensual content generation. The ability to fine-tune generative models using limited data further reduces the barrier to entry for misuse, allowing individuals without advanced technical expertise to produce convincing synthetic media (Chesney and Cytowic, 2019). This tension complicates regulatory and technical responses to deepfakes, as restrictive controls on generative technologies risk constraining legitimate innovation, while inadequate oversight allows harmful applications to proliferate (Chesney and Cytowic, 2019; Westerlund, 2019).

From a forensic perspective, this dual-use nature highlights the limitations of prevention-only strategies and reinforces the need for effective detection mechanisms (Chesney and Cytowic, 2019). Detection systems must therefore be capable of operating independently of the intent behind content creation, instead focusing on identifying artefacts, inconsistencies, or statistical traces that distinguish synthetic media from authentic recordings (Rossler et al., 2019; Dolhansky et al., 2020). Understanding deepfakes as a dual-use technology provides an important context for evaluating both generation methods and detection strategies. It underscores why detection research must prioritise generalisation and resilience to evolving synthesis techniques, rather than relying on assumptions about specific models or controlled deployment scenarios (Rossler et al., 2019; Dolhansky et al., 2020; Cai, 2024). This perspective motivates the examination of deepfake generation techniques in the following section, as well as the detection methodologies reviewed later in this chapter.

2.5 Deepfake Generation Techniques

Deepfake generation techniques can be broadly categorised according to the modality they manipulate: images, audio, or video. While early systems often focused on a single modality, recent advances increasingly integrate multiple streams, complicating forensic analysis. Understanding the mechanisms behind these generation techniques is essential for contextualising the strengths and limitations of current detection approaches (Rossler et al., 2019; Dolhansky et al., 2020; Yi et al., 2023; Cai, 2024).

2.5.1 Image-Based Manipulation

Image-based deepfake generation primarily targets facial appearance through identity replacement, expression transfer, or attribute manipulation (Rossler et al., 2019). Early approaches relied on autoencoder architectures trained to map facial representations between source and target identities, enabling face swapping with relatively limited training data (Dolhansky et al., 2020). These systems commonly employed a shared encoder with identity-specific decoders, allowing facial expressions and pose information to be transferred while preserving identity-related features (Gera, 2018). Subsequent progress was driven by the adoption of Generative Adversarial Networks (GANs), which enabled higher resolution synthesis and more realistic texture generation (Rossler et al., 2019). Advances in training strategies and loss formulation, including stabilisation techniques such as the Two-Time Scale Update Rule, significantly reduced artefacts such as colour inconsistency and boundary distortion, making image-based deepfakes increasingly difficult to detect using handcrafted forensic cues (Heusel et al., 2017; Li et al., 2019). Recent studies have incorporated attention mechanisms and multi-scale discriminator architectures to improve performance under common post-processing operations such as compression and resizing, which can reduce visual differences between synthetic and authentic facial images (Cai, 2024).

2.5.2 Audio-Based Manipulation

Audio-based deepfake generation focuses on synthesising or converting speech to mimic the voice characteristics of a target speaker. Two dominant paradigms underpin these systems: text-to-speech (TTS), which generates speech directly from text, and voice conversion (VC), which transforms the vocal attributes of a source speaker into those of a target speaker while preserving linguistic content (Yi et al., 2023). Early systems relied on statistical parametric models, but recent advances employ deep neural architectures capable of producing highly natural and expressive speech (Korshunov and Marcel, 2018). Modern audio deepfakes leverage end-to-end neural models, including encoder–decoder architectures and diffusion-based approaches, which allow high-fidelity voice cloning from limited reference audio (Shen et al., 2023). These systems can accurately reproduce speaker identity, prosody, and emotional tone, making synthetic speech difficult to distinguish from genuine recordings by both humans and automated verification systems (Korshunov and Marcel, 2018; Yi et al., 2023). The increasing accessibility of such tools has lowered the barrier to misuse, particularly in impersonation and fraud scenarios, posing significant challenges for existing audio authentication mechanisms.

2.5.3 Video-Based Manipulation

Video-based deepfake generation extends beyond static image manipulation by modelling temporal consistency across frames, allowing for realistic facial motion, head pose variation, and synchronisation with speech (Rossler et al., 2019). Early video deepfakes often exhibited temporal artefacts such as flickering, inconsistent lighting, or unnatural motion, which could be exploited by detection systems (Li et al., 2018). However, advances in spatio-temporal modelling have substantially mitigated these weaknesses. Recent video-based techniques integrate facial reenactment, motion transfer, and lip synchronisation models to produce coherent and temporally stable outputs (Dolhansky et al., 2020). The emergence of multimodal generation frameworks further enables joint manipulation of video and audio streams, requiring generators to maintain consistency across visual appearance, speech content, and timing (Cai, 2024). Additionally, neural rendering approaches such as Neural Radiance Fields have been adopted to synthesise realistic talking heads with controllable viewpoints and lighting conditions, further increasing realism and generalisation across scenarios (Guo et al., 2021). These developments significantly complicate forensic analysis, as detectors must now take into account both spatial and temporal cues across multiple modalities.

2.6 Deepfake Detection Methodologies

Deepfake detection methodologies aim to distinguish synthetic media from authentic recordings by identifying artefacts, inconsistencies, or statistical patterns introduced during the generation process. As deepfake generation techniques have evolved, detection approaches have progressed similarly from handcrafted forensic features to data-driven deep learning models. However, despite substantial advances, existing detection systems continue to struggle with generalisation, particularly under real-world conditions (Rossler et al., 2019; Dolhansky et al., 2020).

2.6.1 Image-Based Detection Approaches

Image-based deepfake detection methods focus on identifying spatial inconsistencies within individual frames. Early approaches relied on handcrafted forensic cues, exploiting artefacts such as abnormal eye blinking, colour mismatches, and unnatural facial boundaries that were common in early face-swapping systems (Li et al., 2018). While effective against early deepfakes, these methods lacked robustness as generative models improved. The adoption of convolutional neural networks (CNNs) marked a shift towards learning discriminative features directly from data. Large-scale benchmarks such as FaceForensics++ demonstrated that CNN-based detectors could achieve high accuracy within controlled datasets, often exceeding 0.95 AUC (Rossler et al., 2019). Subsequent research explored frequency-domain representations, showing that GAN-generated images exhibit characteristic spectral artefacts that can be exploited for detection (Li et al., 2019). Despite strong performance in the dataset, image-based detectors exhibit significant performance degradation when evaluated in unseen datasets or under post-processing operations such as compression and resizing (Dolhansky et al., 2020). This suggests that many models overfit to dataset-specific artefacts rather than learning generator-invariant features. Recent work incorporating attention mechanisms and multiscale feature extraction has shown incremental robustness improvements, but image-only detection remains insufficient against modern, high-quality deepfakes (Cai, 2024).

2.6.2 Audio-Based Detection Approaches

Audio-based detection methods aim to identify synthetic speech generated through text-to-speech or voice conversion systems. Early research adapted techniques from automatic speaker verification, using features such as Mel frequency cepstral coefficients and phase-based representations to capture artefacts introduced by speech synthesis (Korshunov and Marcel, 2018). These studies demonstrated that synthetic speech often contains over-smoothed spectral patterns and unnatural phase information. More recent approaches employ deep neural architectures to learn discriminative representations directly from raw or minimally processed audio (Yi et al., 2023). While these models achieve strong benchmark performance, they remain vulnerable to unseen synthesis techniques and domain shifts, mirroring the limitations observed in image-based detection (Korshunov and Marcel, 2018). Furthermore, empirical studies show that human listeners struggle to reliably distinguish real and synthetic speech, reinforcing the need for automated audio deepfake detection systems (Yi et al., 2023).

2.6.3 Video-Based Detection Approaches

Video-based detection approaches extend image-level analysis by incorporating temporal information across frames. Early methods exploited temporal artefacts such as inconsistent head motion, unnatural blinking patterns, and frame-to-frame discontinuities (Li et al., 2018). These cues enabled sequence-level classification using temporal aggregation or recurrent architectures. As generative models improved, many temporal artefacts became less pronounced. Contemporary video detectors therefore rely on spatio-temporal architectures, including 3D convolutional networks and attention-based temporal modelling, to capture subtle motion inconsistencies (Rossler et al., 2019; Dolhansky et al., 2020). Although these methods outperform frame-based detectors in controlled settings, they remain sensitive to compression, frame rate variation, and domain shifts common in real-world video content (Dolhansky et al., 2020).

2.6.4 Multimodal Detection Approaches

The emergence of deepfakes that manipulate both audio and visual streams has driven increasing interest in multimodal detection approaches. These methods jointly analyse facial motion, lip synchronisation, and speech content to identify cross-modal inconsistencies that are difficult for generative models to reproduce perfectly (Yi et al., 2023). Datasets such as FakeAVCeleb and AV-Deepfake1M have enabled systematic evaluation of multimodal detectors under coordinated manipulation scenarios (Cai, 2024). Multimodal detection systems generally demonstrate improved robustness compared to unimodal approaches, particularly in cross-dataset evaluations (Cai, 2024). However, their performance remains constrained by dataset bias, limited availability of synchronised training data, and increased computational complexity, which may hinder real-time deployment.

ResNet3D-18 for Video Feature Extraction

One common approach for video-based deepfake detection employs 3D convolutional neural networks, such as ResNet3D-18, which extends the successful ResNet architecture to three-dimensional convolutions (Tran et al., 2018). ResNet3D-18 is pretrained on large-scale video datasets such as Kinetics-400, which contains diverse human action categories, providing rich visual representations that transfer well to forensic tasks. The 3D convolutions process video as a volume (channels $\times$ time $\times$ height $\times$ width), enabling the model to learn spatio-temporal features that capture both appearance and motion patterns simultaneously. This is particularly valuable for deepfake detection, as synthetic videos often exhibit subtle temporal inconsistencies in facial dynamics that single-frame analysis cannot detect. The pretrained weights from Kinetics-400 provide strong initial representations, reducing the amount of training data required and improving generalisation to unseen manipulation techniques (Crasto et al., 2019).

ResNet18 for Audio Feature Extraction

For audio analysis, researchers have successfully adapted 2D convolutional networks originally designed for image classification to process spectrographic representations of audio signals (Korshunov and Marcel, 2018). ResNet18, pretrained on ImageNet, can be applied to mel-spectrograms treated as single-channel images, leveraging the transfer learning benefits that have made ResNet architectures popular across vision tasks. The network processes the time-frequency representation to identify artefacts introduced by speech synthesis or voice conversion systems. This approach offers a practical balance between detection performance and computational efficiency, as the same architecture family used for video can be applied to both modalities, simplifying the overall system design.

Transformer-Based Cross-Modal Fusion

Recent advances in multimodal deepfake detection have explored transformer architectures to model interactions between audio and visual streams (Cai, 2024). Transformer-based fusion uses self-attention mechanisms to learn fine-grained dependencies between speech content and corresponding visual cues, such as lip movements and facial expressions. The [CLS] token approach, originally developed for BERT-style language models, provides a unified representation that aggregates information from both modalities. This design allows the network to focus on cross-modal inconsistencies—such as mismatches between lip synchronisation and speech—that are characteristic of manipulated content. Compared to simple concatenation or late-fusion approaches, transformer fusion enables earlier and more sophisticated interaction between modalities during feature learning (Cai, 2024).

Generalisation and Cross-Dataset Performance

A central challenge across all detection modalities is poor generalisation to unseen generators and real-world media conditions. Numerous studies report substantial performance drops when detectors trained on one dataset are evaluated on others generated using different synthesis methods (Rossler et al., 2019; Dolhansky et al., 2020). These findings indicate that many detectors rely on superficial cues rather than fundamental properties of synthetic media. Cross-dataset evaluation has therefore become a critical benchmark for assessing practical effectiveness. Recent work emphasises that improving generalisation is more important than achieving marginal gains in closed-set benchmarks (Cai, 2024).

2.7 Gaps in the Literature

Prior work shows that unimodal deepfake detectors often fail to generalise to unseen data, motivating increased interest in audio–visual detection methods (Rossler et al., 2019; Dolhansky et al., 2020; Cai, 2024). However, existing multimodal approaches exhibit several unresolved limitations. Most audio–visual detectors rely on simple feature fusion, typically concatenating independently learned audio and visual representations (Yi et al., 2023). This design limits their ability to capture temporal relationships between speech and facial motion, reducing sensitivity to cross-modal inconsistencies. In addition, many systems remain vision-centric, treating audio as a supplementary rather than an equal source of evidence (Rossler et al., 2019; Cai, 2024). As a result, the contribution of audio information is rarely quantified through systematic ablation or controlled evaluation. Finally, current methods commonly assume that both modalities are reliable, although real-world media often exhibit asymmetric degradation due to noise or compression (Dolhansky et al., 2020).

2.8 Conclusion

This chapter reviewed deepfake generation and detection techniques across image, audio, video, and multimodal domains. Advances in generative models have led to increasingly realistic synthetic media, reducing the effectiveness of early detection approaches based on visible artefacts (Rossler et al., 2019; Dolhansky et al., 2020; Li et al., 2019). Across modalities, a consistent limitation is reduced performance outside controlled benchmarks, particularly when detectors encounter unseen generators or real-world media conditions (Rossler et al., 2019; Dolhansky et al., 2020; Cai, 2024). Multimodal detection offers potential advantages, but remains constrained by design assumptions, limited analysis of modality contributions, and sensitivity to domain shift (Yi et al., 2023; Cai, 2024). These findings highlight the need for detection approaches that prioritise generalisation across modalities and manipulation techniques, motivating the research direction adopted in this dissertation.

Chapter 3: Methodology

3.1 Introduction

This chapter outlines the approach taken to design, develop, and assess a deepfake detection system for audio-visual content utilizing the AV-Deepfake1M++ dataset (Cai, 2025). The aim is to meet research goals systematically while respecting hardware, storage, and computational limits. The work targets content-driven deepfakes, where a real identity is kept but spoken content is altered via audio synthesis and lip synchronisation. Such changes are subtle and localised, making unimodal detection unreliable (Korshunov and Marcel, 2018; Cai, 2024). Therefore, a multimodal strategy is adopted that exploits the mismatches between speech and lip motion.

Details on implementation and experimental findings are provided in the following chapters.

3.2 Research and Development Approach

This study uses a quantitative research approach based on secondary analysis of the AV-Deepfake1M++ dataset (Cai, 2024). Using an existing dataset avoids the cost and practical difficulties of collecting a large-scale audio–visual corpus, which would not be realistic within the scope of this project. To manage development and experimentation, a simple Kanban-based workflow is used. Tasks such as data loading, preprocessing, feature extraction, model integration, and evaluation are treated as separate stages and worked on incrementally. This makes it easier to test individual parts of the system without having to run the full pipeline each time. Each component is first tested using a small subset of video clips to confirm that it behaves as expected. Only after basic performance checks are satisfied is the component integrated into the full pipeline. This step-by-step approach helps reduce errors and unnecessary computation, which is particularly important when working with high-dimensional audio and video data.

3.3 System Architecture and Implementation

3.3.1 Overview of the Proposed System

The main objective of this project is to maximise classification accuracy at the clip level. Although the AV-Deepfake1M++ dataset provides temporal localisation annotations, these are not utilised in the current work due to the high computational cost of frame-level modelling. Given the available hardware and storage constraints, the study focuses on video-level classification.

3.3.2 Dataset Selection and Subset Strategy

The AV-Deepfake1M++ dataset is approximately 1.4 TB in size, which makes full local storage and processing infeasible within the available development environment. To manage this constraint, the study adopts a subset-based strategy that uses selected portions of the training and validation splits. The training subset is used to learn model parameters, while the validation subset supports hyperparameter tuning and early stopping. The TestA and TestB splits are not used during model development.

3.3.3 Audio Feature Extraction Module

For the audio modality, the system uses Wav2Vec 2.0, which is a pretrained self-supervised speech model that extracts feature embeddings directly from raw audio signals (Baevski et al., 2020). Unlike traditional handcrafted features such as MFCCs, Wav2Vec 2.0 learns contextual and prosodic information from speech, making it more suitable for detecting artefacts introduced by modern speech synthesis and voice cloning methods (Yi et al., 2023). Audio is extracted from each video clip and processed separately from the visual stream. The resulting embeddings capture temporal speech characteristics and are forwarded to the fusion module. No explicit temporal localisation is performed at this stage, as the focus of this study is on clip-level classification rather than frame-level analysis.

3.3.4 Visual Feature Extraction Module

The visual stream focuses on the mouth ROI to reduce computational overhead and remove irrelevant facial information. Facial landmark detection is used to locate and crop the lip region of each frame. Visual features are extracted using a lightweight convolutional neural network, such as MobileNetV3, which provides a balance between efficiency and representational capacity (Howard et al., 2019). The extracted features encode spatial characteristics related to lip shape and movement, which are particularly relevant for detecting lip-sync inconsistencies and mouth-related artefacts commonly observed in deepfake videos (Li et al., 2018).

3.3.5 Cross-Modal Fusion and Classification

Audio and visual embeddings are combined using a cross-modal fusion module inspired by the DiMoDif architecture (Discourse Modality-information Differentiation) proposed for the AV-Deepfake1M++ dataset (Cai, 2025). DiMoDif is designed to model fine-grained dependencies between audio and visual streams by learning how speech content (phonemes) aligns with corresponding mouth movements (visemes) over time. This design directly targets content-driven deepfakes, allowing subtle inconsistencies introduced by manipulation to be detected.

Unlike unimodal or late-fusion approaches, DiMoDif encourages interaction between modalities during feature learning, allowing cross-modal inconsistencies introduced by manipulation to be more easily detected (Korshunov and Marcel, 2018). This makes the architecture well suited for audio–visual deepfake detection tasks that prioritise classification accuracy rather than temporal localisation. In this study, the fusion module operates on temporally aligned audio and visual embeddings to learn a shared representation that captures cross-modal coherence. The resulting fused representation is passed to a binary classification head, which outputs the probability that a video being real or manipulated. The temporal boundary detection components of the original DiMoDif framework are not included, as the focus of this project is video-level classification.

3.4 Model Training and Evaluation Strategy

Training is performed on the selected training subset, with early stopping applied based on validation performance to reduce overfitting. The model is optimised using binary cross-entropy loss. Detection accuracy is used as the primary evaluation metric, with precision, recall, and F1-score reported to provide additional insight into classification performance. The TestA and TestB splits are not used for this project.

3.5 Challenges and Limitations

The primary limitation of this study arises from computational and storage constraints, which require the use of dataset subsets rather than full-scale training. Although this limits the model's exposure to the full diversity of manipulations, it allows for meaningful experimentation within realistic hardware boundaries. Another limitation is the exclusion of temporal localisation. Although frame-level manipulation detection could provide deeper insight into model behaviour, such methods require significantly greater computational resources and are therefore deferred to future work. Finally, since the dataset is constructed by third parties, the study relies on the accuracy and completeness of existing annotations. Any bias present in the dataset can influence the performance of the model.

3.5.1 Implementation Deviation from Proposed Methodology

The original methodology (Sections 3.3.3–3.3.5) proposed using Wav2Vec 2.0 for audio feature extraction, MobileNetV3 on mouth ROI for visual features, and DiMoDif fusion for cross-modal interaction. However, the final implementation deviates from this proposal due to practical considerations and available resources.

Audio Encoder: ResNet18 Instead of Wav2Vec 2.0

Wav2Vec 2.0 is a powerful self-supervised speech model that learns contextual representations from raw audio (Baevski et al., 2020). However, it requires significant GPU memory for fine-tuning and has slower inference times compared to traditional CNN-based approaches. In this implementation, ResNet18 pretrained on ImageNet was adapted to process mel-spectrograms as grayscale images. This approach provides comparable detection performance with faster inference and simpler pipeline integration. The mel-spectrogram representation ($1\times128\times63$) is resized to 224×224 to match ResNet's expected input size, leveraging transfer learning from the visual domain.

Video Encoder: ResNet3D-18 Instead of MobileNetV3

The original proposal envisioned using MobileNetV3 on cropped mouth regions (ROI) to reduce computational overhead. In practice, full-frame processing with ResNet3D-18 proved more effective for several reasons. First, ResNet3D-18 maintains temporal information across all 50 frames (2 seconds at 25fps), capturing broader contextual dynamics beyond the mouth region. Second, the Kinetics-400 pretrained weights provide strong initial representations for human action and motion recognition. Third, processing full frames eliminates the need for additional facial landmark detection and lip cropping, simplifying the pipeline and reducing failure points. The output is a 256-dimensional feature vector representing the entire video clip.

Fusion: Transformer/MLP Instead of DiMoDif

The DiMoDif architecture was designed to learn phoneme-viseme alignment by modelling fine-grained temporal dependencies between speech and lip movements (Cai, 2025). This requires precise temporal alignment annotations and significant computational resources. The implemented system uses either a Transformer-based fusion with learnable [CLS] token or a simpler MLP-based fusion (PretrainedFusion). Both approaches concatenate audio and video features and learn a shared representation through self-attention or fully connected layers. While less sophisticated than DiMoDif, these fusion methods still capture cross-modal inconsistencies effectively while being more robust to alignment errors and computationally efficient.

These deviations represent pragmatic adaptations to available computational resources and hardware constraints while maintaining the core principles of multimodal deepfake detection. The system still employs a dual-encoder architecture with separate audio and video branches, cross-modal fusion, and three classification heads (audio authenticity, video authenticity, and joint authenticity).

3.6 Ethical Considerations

The dataset used in this study is publicly available and anonymised, and contains no personally identifiable information. As such, no direct ethical concerns arise from data collection. However, deepfake detection research has broader ethical implications. Although the goal of this work is defensive, the same technologies can be misused to improve synthetic media generation (Westerlund, 2019; Chesney and Cytowic, 2019). Therefore, this study positions its contribution strictly within the context of detection and harm mitigation, without deployment or real-world enforcement claims.

3.8 Conclusion

A lightweight multimodal pipeline combining Wav2Vec 2.0 and MobileNetV3 features via a DiMoDif fusion module was proposed in this chapter. Due to practical constraints, the implementation adapts this architecture to use ResNet18 for audio spectrogram processing, ResNet3D-18 for full-frame video analysis, and simplified Transformer/MLP fusion. Implementation details and results are presented in the following chapter.

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